

Theoretical Formalism: The BMI Lagrangian

To formalize the Bimodal Manifold Interaction (BMI) framework, we define the action, S_{BMI} which accounts for the coupling between the tensor brane and the scalar bulk mode. This represents the departure from standard General Relativity.

The Action (S_{BMI})

$$S_{BMI} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) - \alpha \phi R_{brane} \right]$$

Equations of Motion

Varying the action with respect to the scalar field ϕ yields the modified Klein-Gordon equation:

$$\square \phi + \frac{dV}{d\phi} = -\alpha R_{brane}$$

Physical Significance of the Coupling Constant (α)

The term α represents the coupling between the gravitational curvature R_{brane} and the scalar field ϕ . In the context of GW250114, the observed ECM bias ($\Sigma \approx 4.0$) suggests that the interaction is non-linear at the merger epoch.